

1 Problem 1 (21 points)

Conventions used throughout. The statement’s hint says that a miss penalty is an additional cost *on top of* the hit time, and that a miss is discovered only after the access to that level has been attempted. The hierarchy is therefore modelled by the nested expression

$$\text{AMAT} = HT_{L1} + MR_{L1} P_{L1}, \quad P_{L1} = HT_{L2} + MR_{L2} P_{L2}, \quad P_{L2} = HT_{L3} + MR_{L3} P_{L3}, \quad (1)$$

with $HT_{L1} = 1$, $HT_{L2} = 10$, $HT_{L3} = 50$ CC, and P_{L3} the main-memory access cost. The miss rates supplied in Questions C and D are *local* miss rates, which is exactly what (1) consumes, so no conversion to global rates is needed.

For the memory access itself, the initial access time M “accounts for a single transfer”, so with a block of 128 B and a bus of B bytes only the transfers after the first are charged 10 CC each:

$$P_{L3} = M + 10 \left(\frac{128 \text{ B}}{B} - 1 \right). \quad (2)$$

Finally, the clock is 2.0 GHz, so $1 \text{ CC} = 1/(2.0 \times 10^9 \text{ Hz}) = 0.5 \text{ ns}$.

Question A (3 points)

- (1) **False.** The L1 caches hold only 64 KB each, so with L2 and L3 removed every L1 miss would have to be served by main memory (hundreds of cycles) instead of by a 10- or 50-cycle cache, which raises the average memory access time substantially.
- (2) **True.** L1 must be fast enough to answer in a single clock cycle, so it is built from faster, lower-density SRAM cells with more ports and more decoding logic per byte; L3 is latency-tolerant and can use denser, cheaper cells, so a given capacity costs more in L1 than in L3.
- (3) **False.** Caches are filled on demand, one block at a time, and the program resides in main memory; only the blocks actually being referenced need to be present in L1. Indeed a program larger than 64 KB could never fit in the L1 instruction cache at all, yet still executes correctly.

Question B (6 points)

Block offset. Entries are *Byte* addressable, so the offset field must select an individual byte within a block. With a 128-Byte block,

$$\# \text{offset bits} = \log_2 128 = 7.$$

(The fact that a data word is 64 bit = 8 Bytes does *not* shrink this field: because addressing is per Byte and not per word, the low 3 bits that select a byte inside a word are still part of the address and belong to the offset.)

Index. The L2 cache holds 512 KB = $512 \times 2^{10} = 2^{19}$ Bytes, so the number of blocks is

$$\# \text{blocks} = \frac{2^{19} \text{ B}}{128 \text{ B}} = \frac{2^{19}}{2^7} = 2^{12} = 4096.$$

The cache is 4-way set associative, i.e. 4 blocks per set, so

$$\# \text{sets} = \frac{4096}{4} = 1024 = 2^{10} \implies \# \text{index bits} = \log_2 1024 = 10.$$

Tag. The tag is whatever remains of the 64-bit physical address:

$$\# \text{tag bits} = 64 - \# \text{index} - \# \text{offset} = 64 - 10 - 7 = 47.$$

Answer.

Field	Tag	Index	Block offset
Width (bits)	47	10	7

Check: $47 + 10 + 7 = 64$ bits, as required.

Question C (10 points)

Local miss rates are $MR_{L1} = 5\%$, $MR_{L2} = 40\%$, $MR_{L3} = 20\%$, and the bus is 16 Bytes wide. A 128-Byte block therefore needs $128/16 = 8$ transfers, of which $8 - 1 = 7$ are *additional*, so by (2)

$$P_{L3} = M + 10 \times 7 = M + 70 \quad \text{CC},$$

where M is the initial main memory access time in CCs. Substituting into (1), the requirement $\text{AMAT} \leq 4$ CC reads

$$1 + 0.05 \left[10 + 0.40(50 + 0.20(M + 70)) \right] \leq 4.$$

Unwinding the nesting one level at a time:

$$\begin{aligned} P_{L1} &\leq \frac{4 - 1}{0.05} = 60 \text{ CC}, \\ P_{L2} &\leq \frac{60 - 10}{0.40} = 125 \text{ CC}, \\ P_{L3} &\leq \frac{125 - 50}{0.20} = 375 \text{ CC}, \\ M &\leq 375 - 70 = 305 \text{ CC}. \end{aligned}$$

Conversion. At 2.0 GHz one clock cycle is 0.5 ns, hence

$$M_{\max} = 305 \text{ CC} \times 0.5 \frac{\text{ns}}{\text{CC}} = 152.5 \text{ ns}$$

Check. With $M = 305$ CC: $P_{L3} = 375$, $P_{L2} = 50 + 0.2(375) = 125$, $P_{L1} = 10 + 0.4(125) = 60$, and $\text{AMAT} = 1 + 0.05(60) = 4$ CC exactly, so the bound is attained and is tight.

Question D (2 points)

The miss rates are unchanged and the initial main memory access time carried forward is the maximum value found in Question C, $M = 305$ CC ($= 152.5$ ns). Only the bus width changes, from 16 to 32 Bytes, so a 128-Byte block now takes $128/32 = 4$ transfers, of which $4 - 1 = 3$ are additional. By (2),

$$P_{L3} = 305 + 10 \times 3 = 335 \text{ CC}.$$

Working back up through (1):

$$\begin{aligned}P_{L2} &= 50 + 0.20 \times 335 = 117 \text{ CC}, \\P_{L1} &= 10 + 0.40 \times 117 = 56.8 \text{ CC}, \\AMAT &= 1 + 0.05 \times 56.8 = 3.84 \text{ CC}.\end{aligned}$$

The average memory access time will be 3.84 CCs (equivalently 1.92 ns). As expected, doubling the bus width removes 40 CC of transfer time from the L3 miss penalty and lowers the AMAT below the 4 CC budget of Question C.